

FULL NAME: \_\_\_\_\_

BLOCK: \_\_\_\_\_

## Multiplicity and Factored-Form Graphing

Lesson 3-5 · Period 2 · Teacher Edition

### OBJECTIVES

|                                |                                                                                                                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Math Objective</b>          | Use the multiplicity of each zero to determine whether the graph of a polynomial crosses the x-axis or is tangent to (turns back from) it at each zero.                                                         |
| <b>Language Objective</b>      | Students will use the frame “The zero $x = \underline{\hspace{1cm}}$ has multiplicity $\underline{\hspace{1cm}}$ , so the graph $\underline{\hspace{1cm}}$ the x-axis.” to describe each zero’s graph behavior. |
| <b>Essential Understanding</b> | A factored-form polynomial reveals its zeros directly. The multiplicity of each zero — odd or even — controls whether the graph crosses or is tangent to the x-axis at that zero.                               |

### TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

|                           |                                                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Topic Goals</b>        | Students apply the multiplicity rule to four factored-form polynomials, one per practice item. Every item is drawn from Savvas Lesson 3-5. |
| <b>Essential Question</b> | How does the multiplicity of each zero determine the graph’s behavior at that x-intercept?                                                 |
| <b>Materials</b>          | Pencils; student packet; calculator (optional). No Chromebooks required for Period 2.                                                      |

### [PIN] PRE-FLIGHT NOTE (read before class)

Explore is 5 bank items: Try It 2a → Try It 2b → #14 → #15 → #16. Paced for ~6–7 min per item. If pairs finish early, push them to the Optional Reinforcement: Practice #11, then Practice #12. If pace slows, cut #16 first — the core multiplicity arc (Try It 2a → Try It 2b → #14 → #15) is the minimum viable set.

Tag: bridge from Monday

#### Do Now

DOK: 1

Minutes: 5

#### Teacher does

- Hand out Do Now at door.
- Circulate during 3 min work.
- Cold-call 1 student to share one zero at minute 4.

#### Students do

- Read each factor; write zeros directly.
- Use sentence frame on the sheet.

#### Questions to ask

- What does it mean for  $(x - a)$  to be a factor?
- If  $(x + 6)$  is a factor, what zero does that give you?

**Adult role**

Solo teacher: stand at door, hand out Do Now as students enter. Circulate for ~3 min. Cold-call 1 student to share one zero.

**Do Now (3-5-savvas-q13)**

Sketch the graph of the function by finding the zeros:  $g(x) = (x + 3)(x - 1)(x - 4)$ .

**[CHECK] ANSWER KEY**

Zeros:  $x = -3$ ,  $x = 1$ ,  $x = 4$ . Each factor  $(x + 3)$ ,  $(x - 1)$ ,  $(x - 4)$  equals zero when  $x$  takes the opposite sign of its constant term.

**[TARGET] SENTENCE FRAME (for your answer)**

"The zeros are  $x = \underline{\quad}$ ,  $x = \underline{\quad}$ , and  $x = \underline{\quad}$  because each factor equals zero."

**LAUNCH — Multiplicity Rule (teacher models on screen)**

*Tag: teacher models Example 2*

**Launch****DOK: 2****Minutes: 12****Teacher does**

- Project Savvas Example 2. Think aloud: count each factor; odd  $\rightarrow$  crosses, even  $\rightarrow$  tangent.
- Pause 2 $\times$  for 30-sec partner-checks.
- Do NOT model Try-Its — release to Explore.

**Students do**

- Watch the modeled example silently first.
- Turn-and-talk during each 30-sec pause.
- Copy the Multiplicity Rules card into their page margin if desired.

**Questions to ask**

- Say: "Count the factor. Odd or even?" for each zero.
- What's the difference between "crosses" and "tangent to"?
- Why does the graph turn back at even multiplicity? (Sign doesn't flip.)

**Adult role**

Project Savvas Example 2. Work it aloud with think-aloud voice. Pause 2 $\times$  for 30-sec partner-check: "Tell your neighbor what multiplicity this zero has."

**[BOOK] MULTIPLICITY RULES (keep printed on your page)**

- If a factor appears ONCE or an ODD number of times, the zero has ODD multiplicity  $\rightarrow$  the graph CROSSES the x-axis at that zero.

- If a factor appears an **EVEN** number of times, the zero has **EVEN** multiplicity → the graph is **TANGENT** to the x-axis (touches and turns back) at that zero.
- Example:  $f(x) = (x - 2)^3(x + 1)^2 \rightarrow x = 2$  has multiplicity 3 (crosses);  $x = -1$  has multiplicity 2 (tangent).

### [MIC] TEACHER SCRIPT

1. “Watch me do this first one. I’ll read each factor, count how many times it appears, and decide if the graph crosses or is tangent.”
2. For  $f(x) = (x - 2)^3(x + 1)^2$ : “At  $x = 2$ , the factor appears 3 times — odd — so the graph **CROSSES**. At  $x = -1$ , the factor appears 2 times — even — so the graph is **TANGENT**.”
3. “What’s one thing I just did that I want you to copy?” — cold-call.
4. Do **NOT** do Try-Its for them. Set them up, release to TPS.

### EXPLORE — Think-Pair-Share (35+ min)

*Work with a partner. Use the multiplicity rules above for every problem.*

*Tag: Think-Pair-Share + Practice*

#### Explore

**DOK: 2**

**Minutes: 33**

#### Teacher does

- 4 laps of circulation (~8 min each).
- Lap 1: notice stalls. Lap 2: ask purposeful Qs. Lap 3: check answers. Lap 4: invite 1 pair to share.
- Do not give answers. Press pairs to justify with the rule card.

#### Students do

- Work in pairs through Try-It 2a, 2b, then Practice #14, #15, #16.
- For each zero, write multiplicity + crosses/tangent before sketching.
- Justify with sentence frame aloud to partner before writing.

#### Questions to ask

- What’s the multiplicity of each zero?
- Does the graph cross or is it tangent at each zero?
- How does the end behavior match the leading coefficient + degree?

#### Adult role

Solo teacher: 4 laps of circulation (~8 min each). Lap 1: notice who’s stalling. Lap 2: ask purposeful Qs. Lap 3: check answers. Lap 4: invite 1 pair to share at board.

#### Bank-item labels (in order):

1. Try It 2a
2. Try It 2b
3. Practice #14

4. Practice #15

5. Practice #16

### Try It 2a

Describe the behavior of the graph of the function at each of its zeros:  $f(x) = x(x+4)(x-1)^4$ .

### [CHECK] ANSWER / ANTICIPATED TALK

**Answer key:** Graph crosses the x-axis at zeros 0 and  $-4$  (odd multiplicity 1); turning point at  $x = 1$  (even multiplicity 4).

### Try It 2b

Describe the behavior of the graph of the function at each of its zeros:  $f(x) = (x^2+9)(x-1)^5(x+2)^2$ .

### [CHECK] ANSWER / ANTICIPATED TALK

**Answer key:** Real zeros only: graph crosses at  $x = 1$  (odd multiplicity 5); turning point at  $x = -2$  (even multiplicity 2). Factor  $x^2 + 9$  has no real zeros.

### Practice #14

Find the zeros of the function, and describe the behavior of the graph at each zero:  $f(x) = x^3 - 8x^2 + 16x$ .

### [CHECK] ANSWER / ANTICIPATED TALK

**Answer key:** Factor:  $x(x-4)^2$ . Zeros: 0, 4. The graph crosses the x-axis at 0 (multiplicity 1) and is tangent (touches and turns back) at 4 (multiplicity 2).

### Practice #15

Find the zeros of the function, and describe the behavior of the graph at each zero:  $g(x) = x^3 - x^2 - 25x + 25$ .

### [CHECK] ANSWER / ANTICIPATED TALK

**Answer key:** Factor by grouping:  $x^2(x-1) - 25(x-1) = (x-1)(x^2-25) = (x-1)(x-5)(x+5)$ . Zeros:  $-5, 1, 5$ . The graph crosses the x-axis at  $-5, 1$ , and  $5$  (all simple zeros).

### Practice #16

Find the zeros of the function, and describe the behavior of the graph at each zero:  $f(x) = 9x^4 - 40x^2 + 16$ .

### [CHECK] ANSWER / ANTICIPATED TALK

**Answer key:** Biquadratic: let  $u = x^2 \rightarrow 9u^2 - 40u + 16 = (9u-4)(u-4) = 0 \rightarrow u = 4/9$  or  $u = 4 \rightarrow x = \pm 2/3$  or  $x = \pm 2$ . Zeros:  $\pm 2/3, \pm 2$ . The graph crosses the x-axis at each zero (all simple).

SHARE / SUMMARY (5 min)

## Share / Summary

DOK: 2

Minutes: 5

### Teacher does

- Cold-call 2 pairs to read their sentence-frame sentence.
- Write the rule sentence on board (student-generated).
- Post the Wednesday bridge prompt visibly.

### Students do

- One student per pair reads their sentence aloud.
- Class signals thumbs up/sideways.
- Write the bridge prompt's key term (conjugate pair) in packet margin.

### Questions to ask

- What's a sentence that captures today's rule?
- For Wednesday: if a cubic has zeros 2,  $1 + \sqrt{3}$ , and  $1 - \sqrt{3}$  — what does that say about its coefficients?

### Adult role

Cold-call 2 pairs. Write one sentence on board. Leave the bridge prompt posted for Wednesday Do Now.

### Sentence frames

Pair share: "For  $f(x) = \underline{\hspace{1cm}}$ , the zero  $x = \underline{\hspace{1cm}}$  has multiplicity  $\underline{\hspace{1cm}}$ , so the graph  $\underline{\hspace{1cm}}$  the x-axis at that zero."

Whole class: one pair at random reads their sentence. Class signals thumbs up/sideways.

### [SEARCH] BRIDGE PROMPT (post for Wednesday Do Now)

"A cubic has zeros 2,  $1 + \sqrt{3}$ , and  $1 - \sqrt{3}$ . What is the nature of its coefficients — rational, irrational, or complex? Why?" (Directly prepares Topic 3 Assessment Q#2.)

## EXIT TICKET

## Exit Ticket

DOK: 1-2

Minutes: 2

### Teacher does

- Collect at door as students exit.
- Scan for confusion flags; plan tomorrow's Do Now emphasis accordingly.

### Students do

- Complete 3 sentence stems.

- Be honest about confusion — this is not graded.

#### Questions to ask

- Did students write a complete sentence for #1?
- Did anyone flag confusion in #3?

#### Adult role

Collect at door. Scan flagged items tonight. No formal grade.

#### [CLIPBOARD] EXIT TICKET — Student Form

Today I learned that \_\_\_\_.

The rule I want to remember is \_\_\_\_.

One thing I'm still unsure about is \_\_\_\_.

#### OPTIONAL REINFORCEMENT (not collected)

*If you finish early or want extra practice before Wednesday. Answers on Desmos or ask for a check.*

#### Optional #1 (Savvas Practice)

At what points do the graphs of  $f(x) = x^3 - 2x^2 - 16x + 20$  and  $g(x) = -12$  intersect?

#### [CHECK] ANSWER / ANTICIPATED TALK

**Answer key:** Solve  $x^3 - 2x^2 - 16x + 20 = -12 \rightarrow x^3 - 2x^2 - 16x + 32 = 0 \rightarrow (x - 2)(x + 4)(x - 4) = 0$ .  
Intersection points:  $(-4, -12)$ ,  $(2, -12)$ ,  $(4, -12)$ .

#### Optional #2 (Savvas Practice)

Sketch the graph of the function by finding the zeros:  $f(x) = 3x^3 - 9x^2 - 12x$ .

#### [CHECK] ANSWER / ANTICIPATED TALK

**Answer key:** Factor:  $3x(x + 1)(x - 4)$ . Zeros:  $x = -1, 0, 4$ . Cubic with positive leading coefficient, three simple zeros (graph crosses at each).

### **[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS**

- Provide the Multiplicity Rule card as a sticker/cut-out on their desk.
- Allow color-coding each zero on their factored form before answering.
- Accept verbal sentence-frame completions in lieu of written.

### **[GLOBE] SUPPORTS FOR ELL STUDENTS**

- Sentence frames printed on student packet (don't skip).
- Word card: "multiplicity" = how many times a factor appears.
- "crosses" = goes through; "tangent" = touches and turns back (Savvas terminology).
- Pair ELL students with English-dominant partners for TPS whenever possible.

### **[CLOCK] PERIOD F EXTENSION (65-min class)**

Period F has an extra 10 min. Use Explore +10 min (not Launch +10).

Suggested add: after Practice #16, pair students do Practice #11 or #12 from the Optional Reinforcement page.

Do NOT add a second Launch example — keeping the rhythm simple.